

Data Mining: Concepts & Techniques

By

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ASSOCIATION RULE MINING

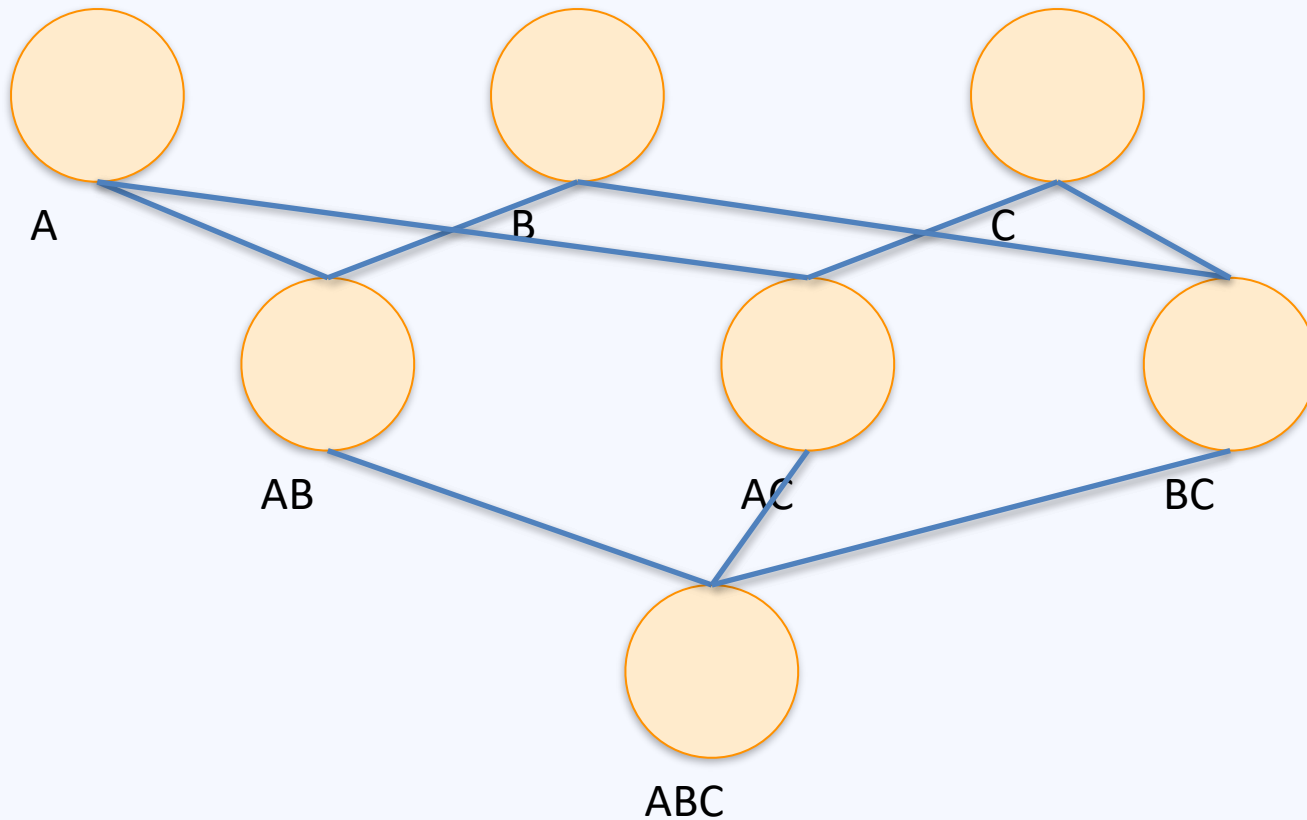
Association Rules — Core Idea

- Define an association rule $X \Rightarrow Y$ over itemsets with measures that reflect frequency and reliability.
- Support $s(X \Rightarrow Y)$: fraction of transactions containing $X \cup Y$; filters rare patterns.
- Confidence $c(X \Rightarrow Y)$: $P(Y|X)$; estimates rule reliability given X .
- Lift = $c / s(Y)$: >1 suggests positive correlation; <1 suggests substitution.
- Typical workflow: find frequent itemsets $\rightarrow$ generate high-confidence rules.

Apriori — Intuition & Properties

- Downward-closure (anti-monotonicity): any superset of an infrequent itemset is infrequent.
- Level-wise search: enumerate candidate k -itemsets from frequent $(k-1)$ -itemsets (join & prune).
- Prune candidates whose any $(k-1)$ -subset is not frequent, reducing scans and counts.
- Compute supports via repeated passes over the database until no new frequent itemsets emerge.

Apriori — Candidate Lattice (Toy Example)



Join: $L_1 \times L_1 \rightarrow C_2$; Prune: drop candidates with infrequent subsets.

Apriori — Worked Example

- Toy DB: T1 {A,B,C}, T2 {A,C}, T3 {A,B}, T4 {B,C}, T5 {A,B,C}; minsup = 60%.
- L1 = {A}(80%), {B}(80%), {C}(60%).
- C2 from L1: {A,B},{A,C},{B,C}; all have 60% support → L2 = all three.
- C3: {A,B,C} has 40% < minsup → stop; generate rules from L2 itemsets.

FP-Growth — Key Steps

- Avoids candidate generation by compressing DB into an FP-Tree using frequency-ordering.
- Two scans: (1) compute frequent 1-items; (2) build FP-Tree with header table and node-links.
- Mine conditional pattern bases for each item (bottom-up) and build conditional FP-Trees.
- Pattern growth combines suffix items with prefix paths to enumerate frequent itemsets.

Scan Database

- Count item frequencies.
- Remove infrequent items.
- Sort frequent items in descending order.

Build FP-Tree

- Use sorted transactions to build a compact prefix tree.
- Share common prefixes.
- Maintain a header table linking similar items.

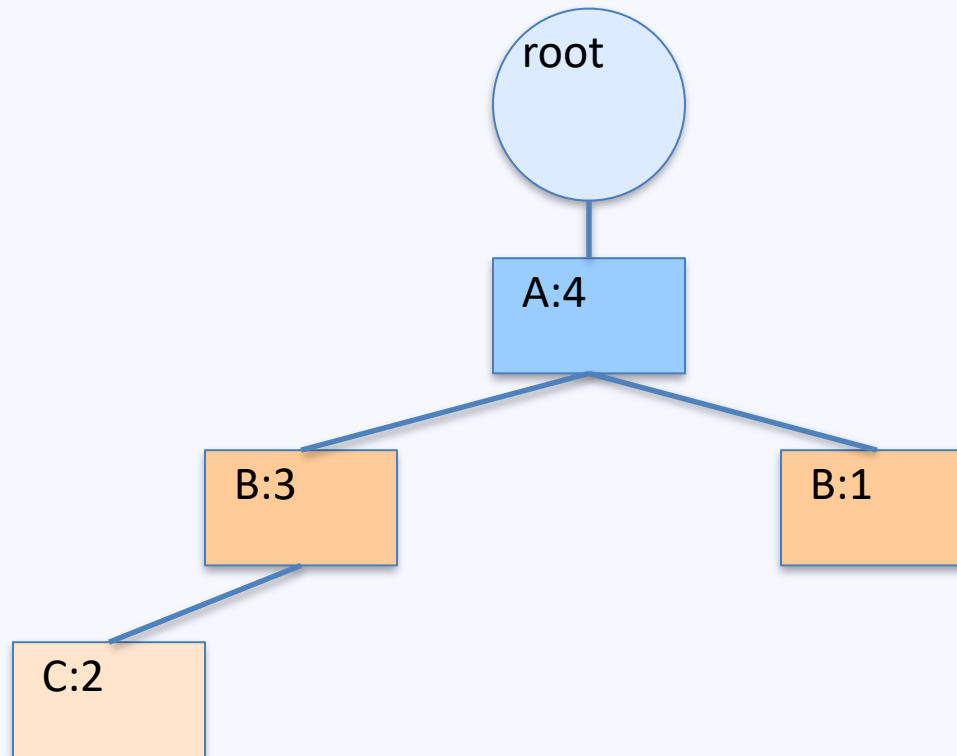
Mine FP-Tree

- For each item (starting from the least frequent):
 - Build **conditional pattern base** (prefix paths).
 - Build **conditional FP-tree**.
 - Recursively mine for frequent itemsets.

Generate Frequent Itemsets

- Combine base item with patterns from recursive mining.

FP-Tree — Schematic (A>B>C order)



Header table links items (A,B,C). Mine conditional bases to grow patterns.

Rule Generation & Postprocessing

- From each frequent itemset F , generate non-empty proper subset A and rule $A \Rightarrow F-A$.
- Prune rules failing minconf; optionally rank by lift/conviction for actionability.
- Closed/maximal itemsets reduce redundancy: closed preserves exact supports; maximal speeds you up.
- Quantitative/multilevel rules handle hierarchies (e.g., milk $\rightarrow$ dairy).

for each frequent itemset L :

- Generate all non-empty subsets of L .

For each subset $S \subset L$:

- Rule: $S \Rightarrow (L - S)$
- Calculate:
 - **Support** = support(L)
 - **Confidence** = support(L) / support(S)
 - **Lift** = confidence / support($L-S$)

Keep rules that meet thresholds:

- **Minimum Confidence**
- Optionally: Minimum **Lift** or **Conviction**

CLASSIFICATION & PREDICTION

Supervised Learning Pipeline

- Data preparation: feature engineering, handling missing values, class imbalance (resampling/weights).
- Model learning: choose algorithm & hyperparameters; use cross-validation for selection.
- Evaluation: confusion matrix, precision/recall, F1, ROC-AUC (classification); MAE/RMSE (prediction).
- Deployment: threshold selection, calibration, monitoring for drift.

Evaluation Metrics (Classification)

- Accuracy = $(TP+TN)/(All)$ can mislead on imbalanced data.
- Precision measures purity of positives; Recall measures completeness; F1 balances both.
- ROC curve plots TPR vs FPR across thresholds; AUC summarizes ranking quality.
- For rare positives, PR curves are more informative than ROC.

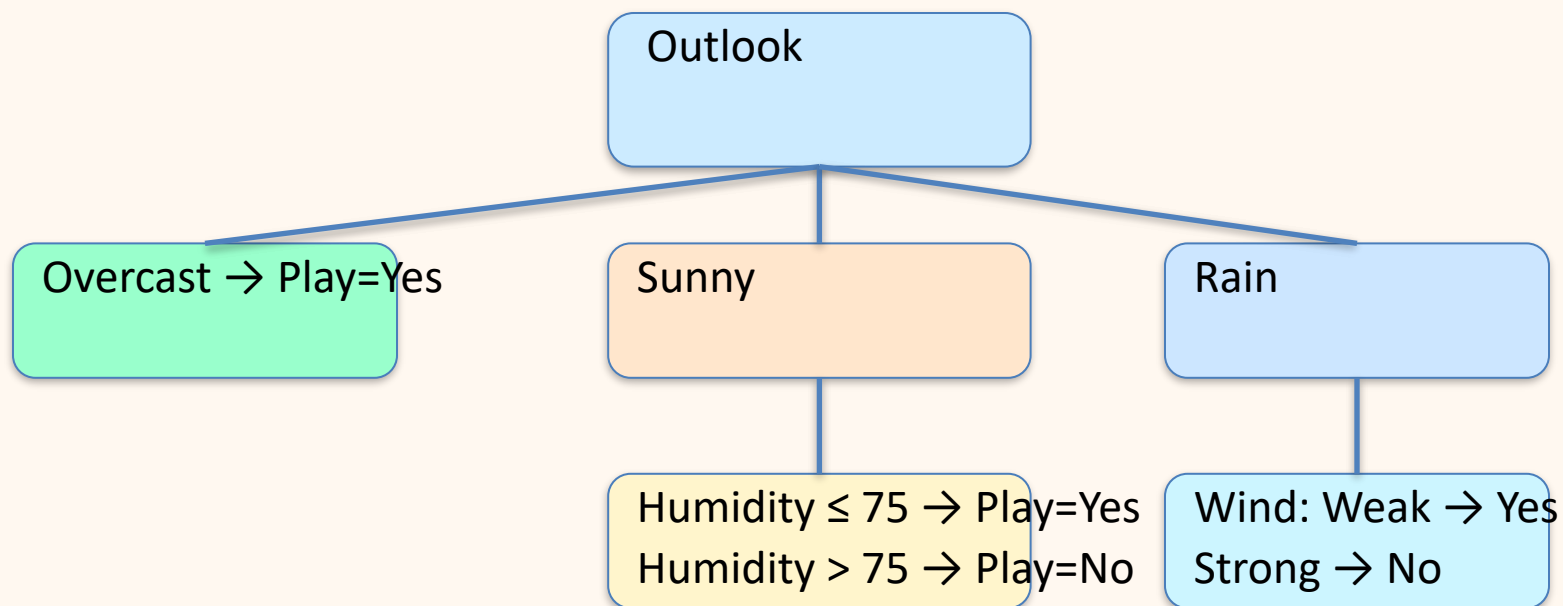
Decision Trees — Concepts & Splits

- Recursive partitioning of feature space into axis-aligned regions.
- Split criteria: Information Gain (ID3), Gain Ratio (C4.5), Gini Impurity (CART).
- Handling numeric features via threshold splits; categorical with multiway or binary splits.
- Missing values via surrogate splits or proportional distribution.

Decision Tree — Basic Algorithm

- Start with all training data at the root; if pure or small, make a leaf with majority class.
- For each attribute, evaluate split quality; choose the best; partition data accordingly.
- Recurse on children; stop by purity, min samples, max depth, or minimal gain criteria.
- Post-pruning with validation (reduced-error) or cost-complexity pruning can improve generalization.

Decision Tree — PlayTennis (Illustrative)



ID3-style structure: root=Outlook; thresholds for numeric attributes such as Humidity.

Bayesian Classification — Essentials

- Bayes decision rule: choose class C maximizing posterior $P(C|x) \propto P(x|C)P(C)$.
- Naïve Bayes assumes conditional independence among features given the class.
- Text classification: Multinomial/Bernoulli NB on word counts/binaries; Gaussian NB for continuous data.
- Use Laplace/add- α smoothing to avoid zero likelihoods for unseen feature values.

$$P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)}$$

$\psi(y)$

Prediction (Regression) — Overview

- Model continuous targets: linear regression, regularized models (Ridge/Lasso), SVR, neural nets.
- Check assumptions for inference; for pure prediction, focus on generalization error.
- Evaluate with MAE/RMSE; inspect residual plots for bias/heteroscedasticity.
- Feature scaling and interaction terms can improve fit for certain models.

OTHER CLASSIFICATION METHODS

k-Nearest Neighbor (k-NN)

- Instance-based: store training data; classify by majority vote among k nearest points.
- Distance metrics: Euclidean/Manhattan for numeric; Cosine for high-dimensional sparse text.
- Hyperparameters: k, distance weighting, tie-breaking; scale features to avoid dominance.
- Curse of dimensionality: distances concentrate; use feature selection or dimensionality reduction.

Support Vector Machines (SVM)

- Max-margin classifier: separates classes with maximum geometric margin.
- Soft-margin with penalty C handles non-separable data via hinge loss.
- Kernel trick maps to higher dimensions: linear, RBF (γ), polynomial (degree).
- Decision function uses support vectors only, improving robustness.

Ensemble Methods

- Bagging: bootstrap resampling; average/vote to reduce variance (Random Forest adds random feature subspaces).
- Boosting: sequential learners focus on errors; AdaBoost and Gradient Boosting improve margins.
- Stacking: meta-learner combines heterogeneous models using out-of-fold predictions.
- Often outperform single models; trade interpretability for accuracy.

Neural & Rule-Based Classifiers

- MLP: fully connected layers with nonlinear activations; trained by backpropagation and SGD/Adam.
- Regularization: dropout, weight decay, early stopping; normalization accelerates training.
- Rule-based learners (e.g., RIPPER, FOIL) produce IF–THEN rules; useful when transparency is required.
- Associative classification builds rules from frequent patterns linked to class labels.

CLUSTER ANALYSIS

Clustering — Concepts & Pipeline

- Goal: group objects so intra-cluster similarity is high and inter-cluster similarity is low.
- Preprocess: standardize features, remove outliers, reduce dimensionality if needed.
- Choose algorithm based on shape/scale/noise: partitioning, hierarchical, density-, grid-, or model-based.
- Validate clusters: internal indices (Silhouette), external when labels exist (ARI/NMI).

Similarity & Distance Measures

- Numeric features: Euclidean/Manhattan/Minkowski; cosine similarity for vectors.
- Binary variables: Jaccard (asymmetric) vs. simple matching (symmetric).
- Categorical/ordinal: encodings or specific measures; mixed-type: Gower distance.
- Scaling is critical—unscaled features can dominate distance-based clustering.

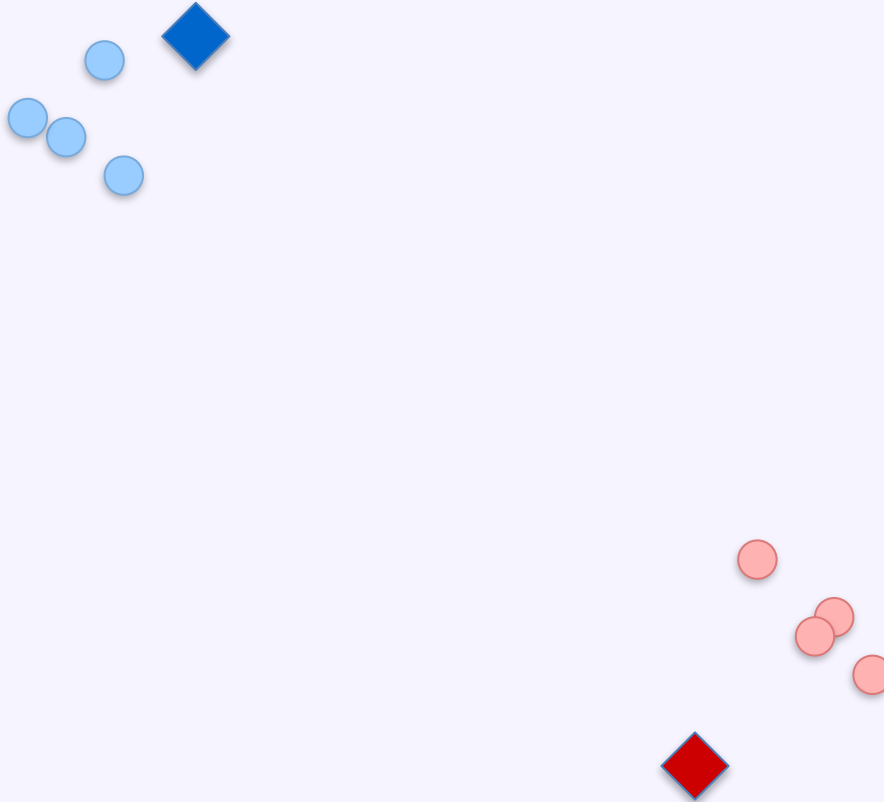
Types of Data in Clustering

- Interval/ratio-scaled variables: continuous measurements (e.g., height, income).
- Binary variables: symmetric (both states equally important) or asymmetric (only presence counts).
- Categorical/nominal and ordinal variables: require appropriate encodings or tailored dissimilarities.
- Mixed-type records: combine measures (e.g., Gower); handle missingness carefully.

k-Means — Algorithm & Objective

- Objective: minimize within-cluster sum of squares (SSE).
- Algorithm: initialize k centroids (random/k-means++), assign points, recompute centroids, iterate.
- Converges to a local minimum; initialization affects results; repeat with different seeds.
- Best for spherical, similarly sized clusters; sensitive to outliers.

k-Means — Stepwise Illustration



Assign points to nearest centroid (diamonds) → recompute centroids as means → iterate.

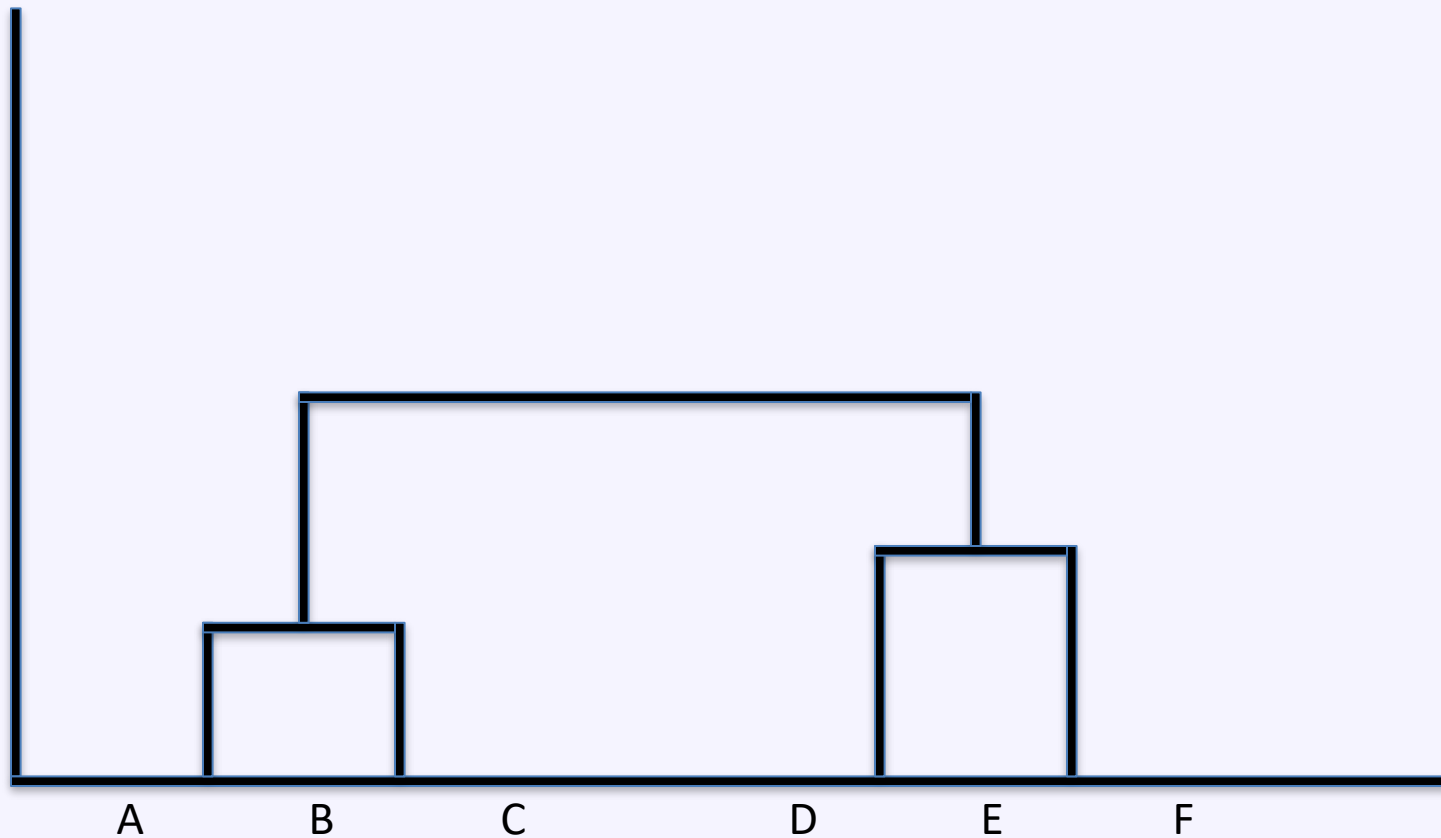
k-Medoids (PAM) & Variants

- k-Medoids uses actual data points as centers to reduce sensitivity to outliers.
- PAM iteratively swaps medoids with non-medoids to improve total dissimilarity.
- CLARA and CLARANS scale PAM via sampling or randomized search for large datasets.
- Useful when distance is non-Euclidean or features include categorical variables.

Hierarchical Clustering — AGNES/DIANA

- Agglomerative (bottom-up): start with singletons and merge clusters based on linkage.
- Divisive (top-down): start with all items and split clusters iteratively.
- Linkage choices: single (min), complete (max), average, Ward's (variance minimization).
- Produces a dendrogram; no need to pre-specify k ; cut tree at desired level.

Dendrogram — Schematic

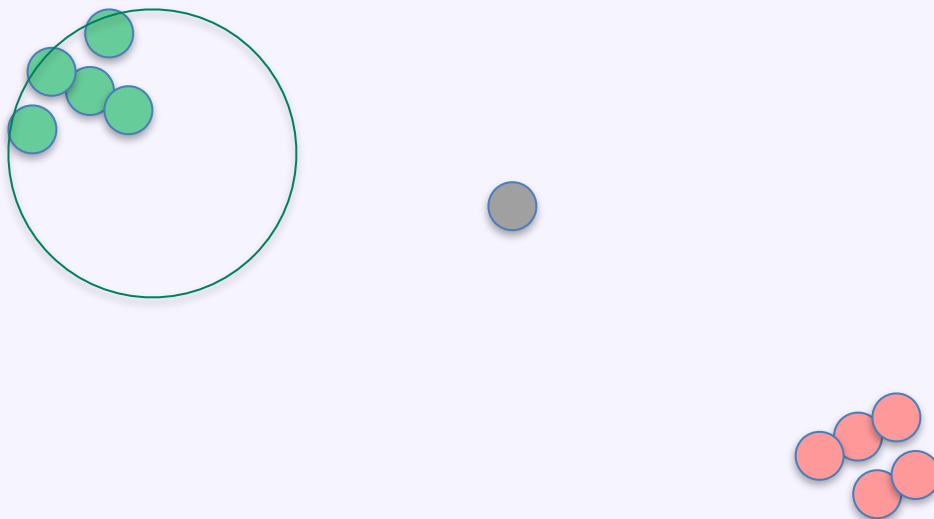


Cut at a chosen height to obtain the number of clusters desired.

Density-Based Clustering — DBSCAN

- Parameters: ϵ (neighborhood radius) and MinPts (minimum points to form a dense region).
- Point types: core ($\geq \text{MinPts}$ in ϵ), border (reachable from a core), noise (not density-reachable).
- Discovers arbitrarily shaped clusters; robust to outliers; no need to predefine k .
- Sensitive to scale: choose ϵ via k -distance plot; standardize features.

DBSCAN — Core, Border, and Noise



ϵ -neighborhood shown as circle; points inside count toward MinPts to form core clusters.

Grid- & Model-Based Methods

- Grid-based: partition space into cells; STING stores statistical summaries per cell; CLIQUE finds dense subspaces.
- Model-based: Gaussian Mixture Models with EM alternate assignment (E) and parameter update (M).
- BIC/AIC for model selection; BIRCH and CURE help large-scale clustering with summaries and robust representatives.

Cluster Validity & Choosing k

- Internal indices: Silhouette combines cohesion and separation; elbow method inspects SSE vs k.
- External indices: Adjusted Rand Index (ARI), Normalized Mutual Information (NMI) when labels exist.
- Stability across resamples and gap statistic provide additional guidance.
- Combine multiple criteria; choose k that balances interpretability and performance.

Linear Regression — Algorithm Details

- Ordinary Least Squares minimizes squared residuals; closed-form $\hat{\beta} = (X^T X)^{-1} X^T y$.
- Regularization combats overfitting: Ridge (L2) shrinks coefficients; Lasso (L1) induces sparsity.
- Evaluate with RMSE/MAE; examine residuals for nonlinearity or heteroscedasticity.
- Feature scaling and polynomial terms can capture curvature.

Naïve Bayes Example (Spam)

- Vocabulary {offer, buy, meeting}. Count occurrences per class and compute likelihoods with add-1 smoothing.
- Given email 'buy offer', compute log-posteriors $\log P(C) + \sum \log P(\text{word} | C)$ for Spam vs Ham.
- Prediction selects higher posterior; illustrates how NB leverages word frequencies.
- NB is fast and strong on high-dimensional sparse text.

ASSESSMENT



MCQs (1–3)

- 1) In Apriori, which property enables pruning of candidate itemsets?
A) Upward-closure B) Downward-closure C) Convexity D) Anti-support
- 2) Which split criterion is used by CART?
A) Information Gain B) Gain Ratio C) Gini Impurity D) Entropy
- 3) In Naïve Bayes, Laplace smoothing primarily prevents:
A) Overfitting B) Zero probabilities C) Class imbalance D) High variance
- 4) Which method can discover arbitrarily shaped clusters without specifying k?
A) k-Means B) PAM C) DBSCAN D) Ward's
- 5) Random Forests primarily reduce which issue of single decision trees?
A) Bias B) Variance C) Class overlap D) Missing data

Summary & References

- We covered: association rules (Apriori, FP-Growth), classification (trees, Bayes, SVM, k-NN, ensembles), and clustering (k-means, hierarchical, DBSCAN, grid/model).